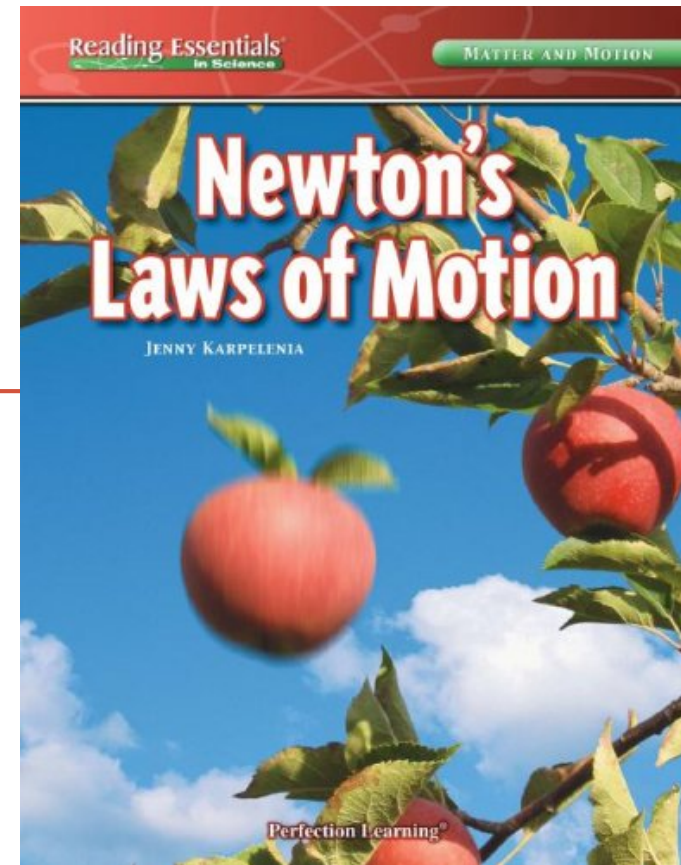


LECTURE 2

DYNAMICS

OF PARTICLE



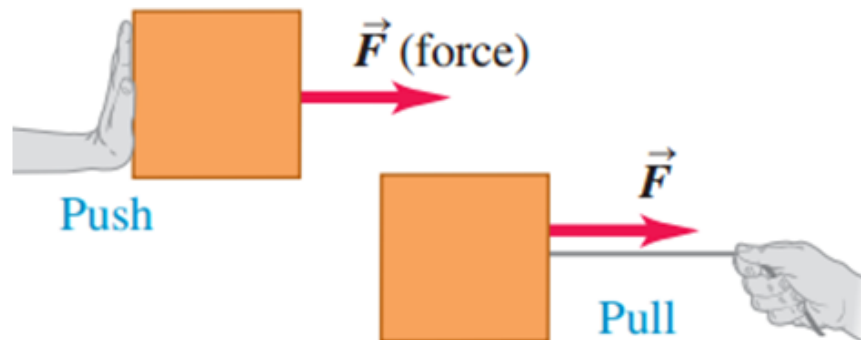
Force and Mass

1. Force $\vec{F}$: characterizes the interactions between objects.

- Is the interaction between two things or things with surroundings.
- Can be repulsive or attractive.
- Is a vector.

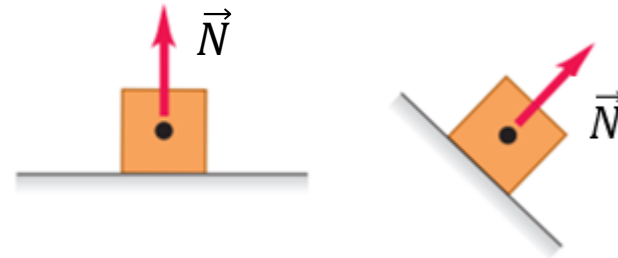
Superposition principle of forces:

$$\vec{F}_1 + \vec{F}_2 = \sum \vec{F}$$

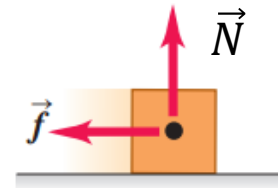


Contact forces

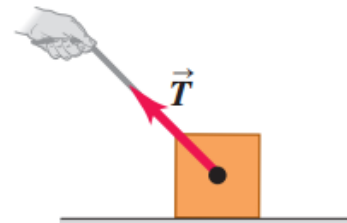
Normal force $\vec{N}$:



Frictional force $\vec{f}$:



Tension of string $\vec{T}$:



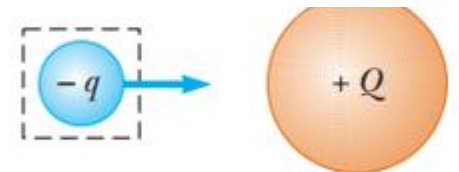
Force fields

Gravitational force



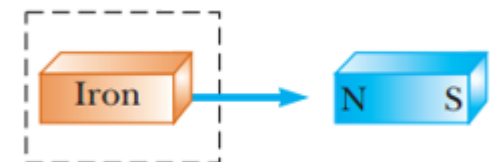
(d)

Electrostatic force



(e)

Magnetic force



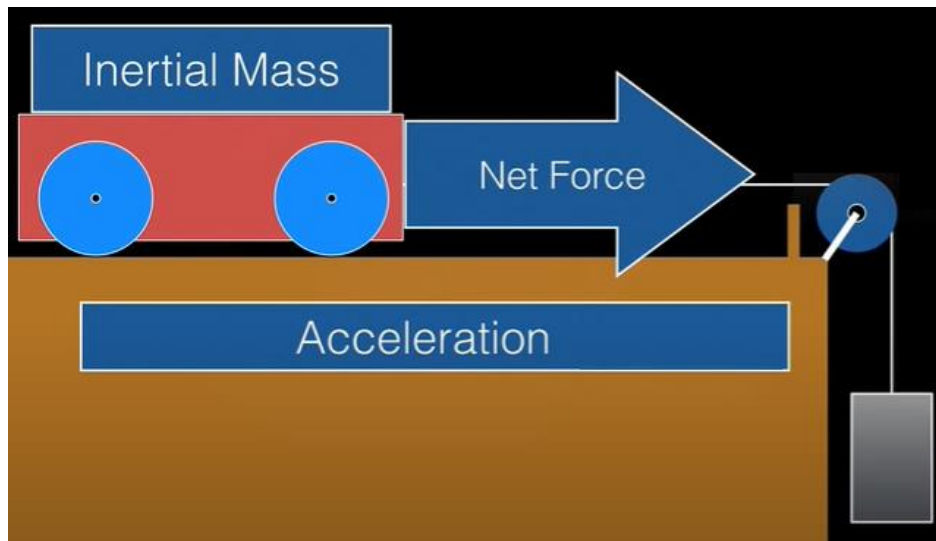
(f)

Force and Mass

2. Mass m :

- Inertial mass: characterizes the inertial degree of the object, according to Newton's second law.
- Gravitational mass: characterizes the gravitationally attracting degree of an object to other objects, according to the law of universal gravitation.

Mass, weight, gravitational force?



Three Newton's laws of motion

Newton's 1st law: <https://www.youtube.com/watch?v=adLj6kygwds> (inertial concept)

In an **Inertial Reference frame**, if no net force acts on an object, it will be at rest or moves with constant velocity

$$\sum \vec{F} = 0 \Rightarrow \vec{v} = \overrightarrow{\text{const}}$$

Newton's 2nd law:

In an **Inertial Reference frame**, the net force acting on an object is equal to the product of the object's mass and its acceleration

$$\vec{F} = m\vec{a}$$

(let's consider the same forces, different masses, how about accelerations?)

(more on acceleration) <https://www.youtube.com/watch?v=vxFYfumAAIY>

Newton's 3rd law:

If A exerts on B a force $\vec{F}$, B exerts on A a force $\vec{F}'$.
 $\vec{F}$ and $\vec{F}'$ are action and reaction force pair

$$\vec{F}' = -\vec{F}$$

<https://www.youtube.com/watch?v=6SA8-E2X7Xg>

Force is the Cause of Changes in Motion

- Force *does not* cause motion.
- We can have motion in the absence of forces, as described in Newton's first law. **Force is the cause of *changes* in motion, as measured by acceleration.**

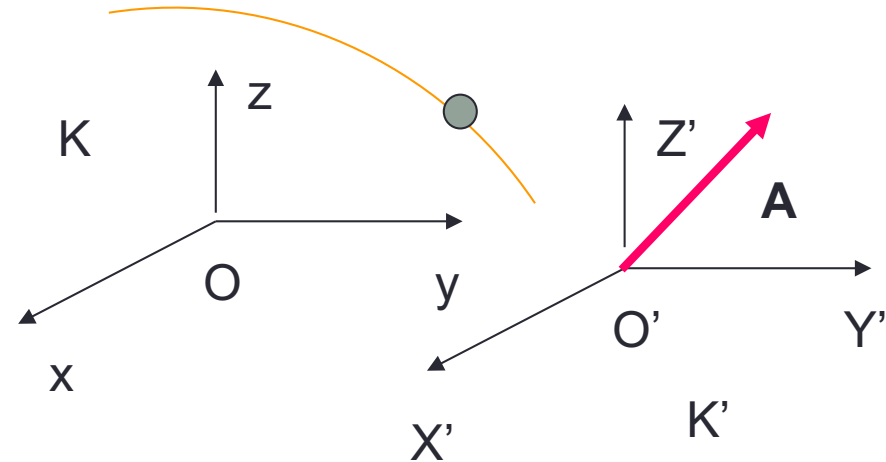
Inertial frames of reference

Non-inertial frames of reference

- Frames in which Newton's laws are valid are called inertial frames of reference.
- The ground (Earth) is an approximate inertial frame.
- Any frame moving at constant velocity relative to an inertial frame is also an inertial frame.
- Any frame moving with acceleration relative to an inertial frame is a non-inertial reference frame

Non-inertial frames – 1

- Observe a particle's motion from two frames O and O' .
- O' has an acceleration $\mathbf{A}$ relative to O .
- Relation between the particle's accelerations relative to two frames:



$$\vec{a} = \vec{a}' + \vec{A}$$

How the behavior of object compared to the reference systems?

<https://www.youtube.com/watch?v=Vk6qZenUGKQ>

Non-inertial frames – 2

- Suppose that frame O is inertial, then from Newton's 2nd law and the above relation we get:

$$m\vec{a} = \vec{F}_{net} = m\vec{a}' + m\vec{A}$$

$$m\vec{a}' = \vec{F}_{net} - m\vec{A}$$

- The latter shows that Newton's 2nd law does not hold in frame O'.
- A frame in which Newton's laws does not hold is called a non-inertial frame.
- Any frame accelerating with respect to an inertial frame is a non-inertial frame.

Dynamics Equation in non-inertial frame of reference

$$m\vec{a}' = \vec{F}_{net_real_force} + \vec{F}_{inert}$$

Inertial force: $\vec{F}_{inert.} = -m\vec{A}$

Example 1 (Newton's second law)

An object of mass $m = 0.5$ kg moves in a straight line. Know that the dependence of the traveled distance s on time t given by the equation: $s = A - Bt + Ct^2 - Dt^3$, in which $C = 5$ m/s² and $D = 1$ m/s². Find the force acting on the object after the first second of motion.

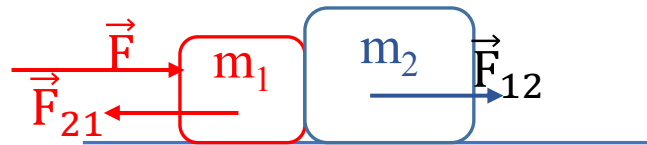
Method for solving problems using Newton's laws

1. Draw the force vectors for each object.
2. Write the equations of the force vectors applying Newton's second law.
3. Select an inertial reference frame.
4. Project the equations of the force vectors on the axes of the reference frame.
5. Solve for unknown variables.
6. Check to see whether the obtained results have correct units and are reasonable.

Example 2 (Newton's third law)

Given a system as shown in the figure. Push the system with a force $\vec{F}$ in the horizontal direction. Ignore all friction. Calculate the interaction force between m_1 and m_2 when:

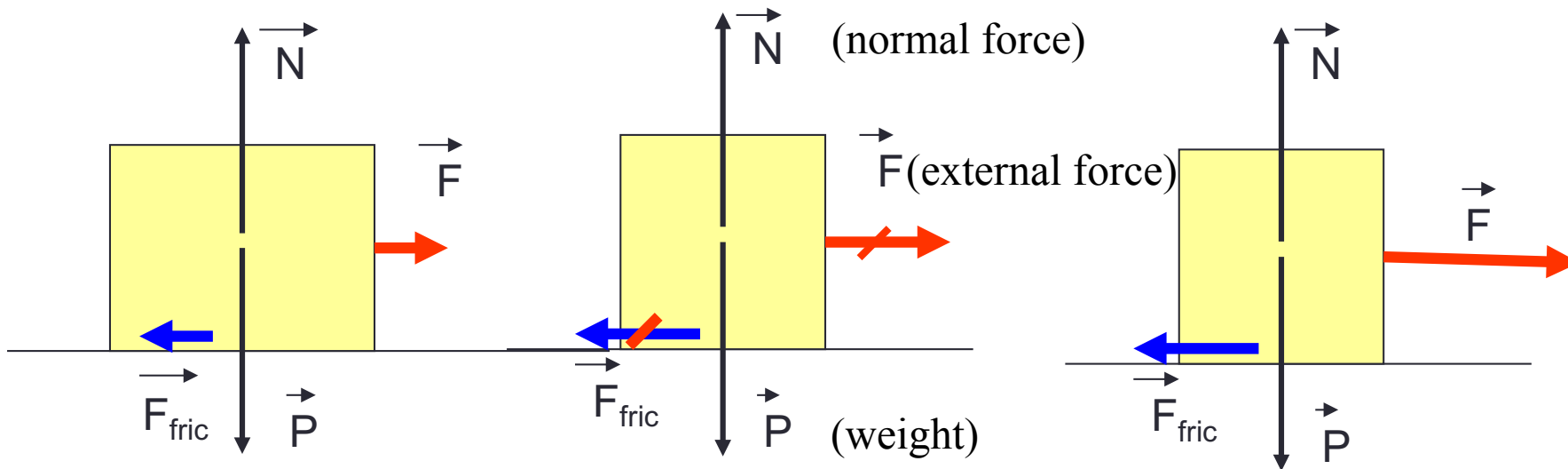
- a) Force $\vec{F}$ is located on m_1 .
- b) Force $\vec{F}$ is located on m_2 .



FRICTION FORCES

1. **Friction force** (https://www.youtube.com/watch?v=_LSevw1sfpk)
 - Static Friction
 - Kinetic friction
2. **Drag forces**
(Types of friction) <https://www.youtube.com/watch?v=qN0V0NXV3Kw>

FRICTION



Apply a force F , object doesn't move, there is a static friction force that balances F

$$F_{\text{fric}} = F, \vec{F}_{\text{fric}} = -\vec{F}$$

Increase force F , object doesn't move, F_{fric} also increases and balances F

Continue to increase F , object begins to move, at that moment $F_{\text{fric}} = F_{\text{fric_max}} = \mu_s N$

When object moves, friction is kinetic friction and constant, even if we continue to increase F . $F_{\text{fric}} = \mu_k N$

μ_s = static friction coefficient

μ_k = kinetic friction coefficient

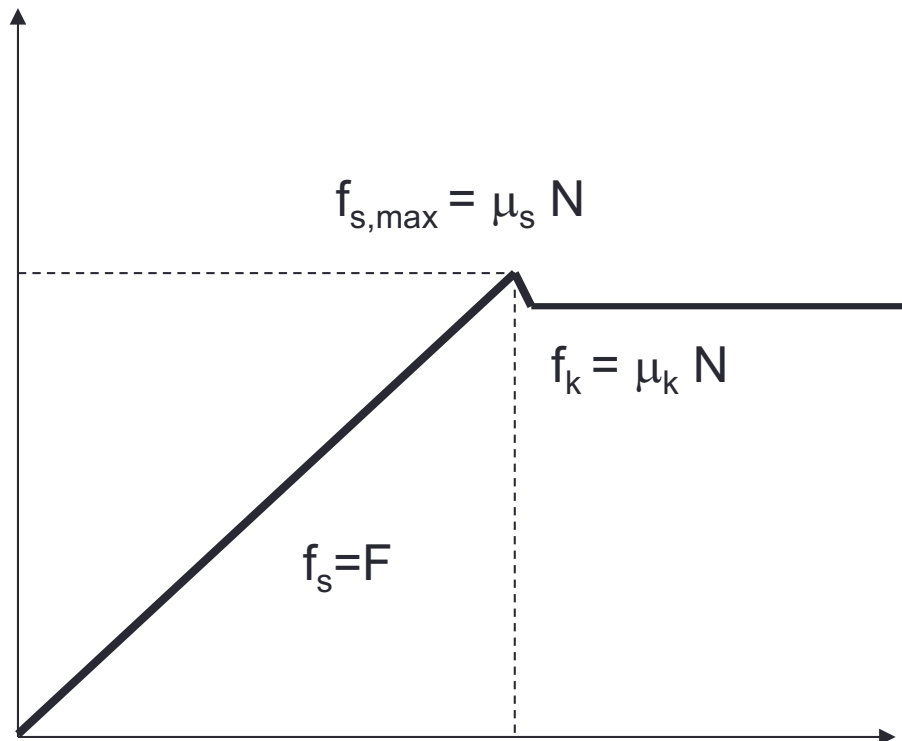
N: normal force

$$\mu_s \geq \mu_k$$

Approximate Value of Frictional Coefficients

Material	μ_s	μ_k
Steel on steel	0.7	0.5
Glass on glass	0.9	0.4
Teflon on Teflon	0.04	0.04
Rubber on concrete (dry)	1.0	0.8
Rubber on concrete (wet)	0.3	0.25

Friction force f



Object at rest

Object
moves

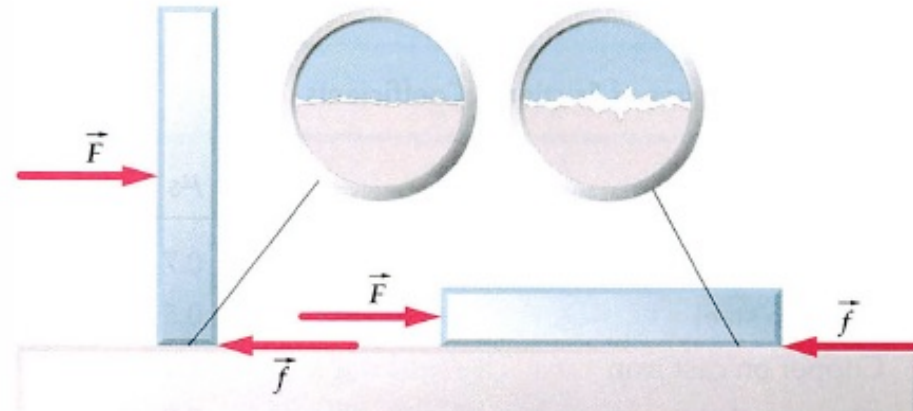
F

Applied force

Object at rest: $f_s \leq \mu_s N$

Object moves: $f_k = \mu_k N$

Friction Explained

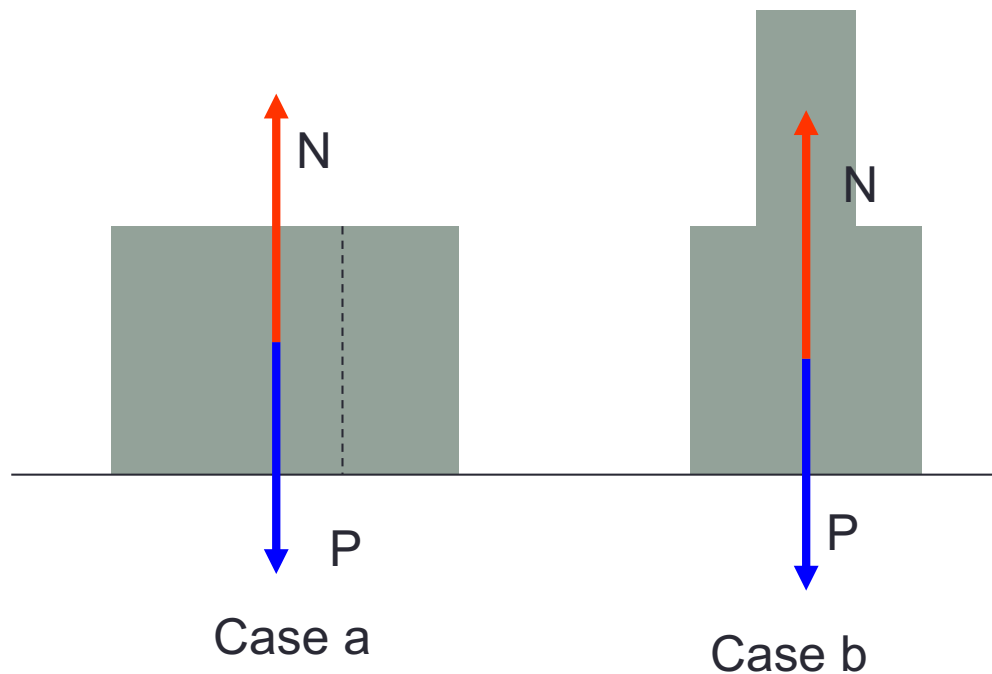


Friction is caused by the attraction of molecules between two surfaces in close contact.

The attraction has the nature of electromagnetic force, the same as the molecule bonding that holds an object together.

Friction coefficients **depend on the nature of surfaces** in contact.

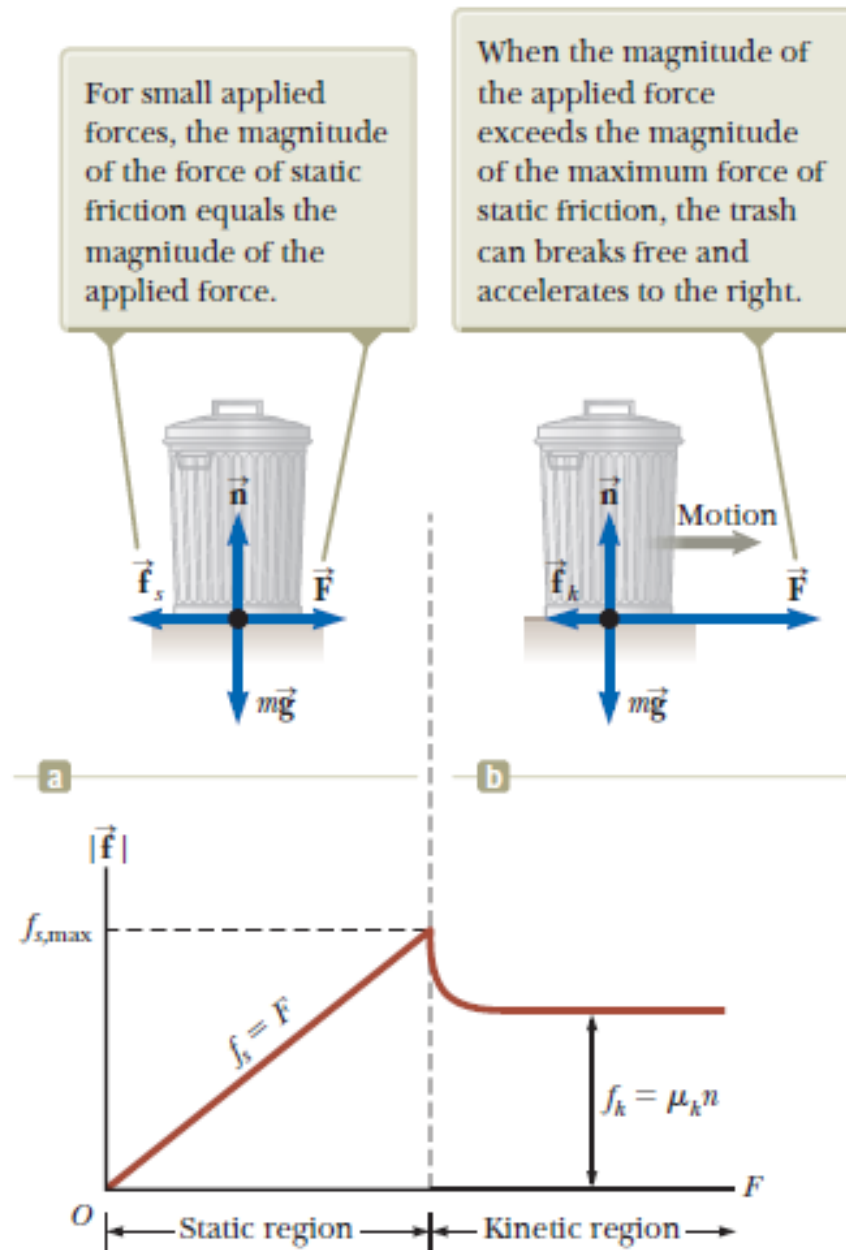
Friction coefficients **do not depend on the area** of contact surfaces



$$f_{s,\max} = \mu_s N$$

$$f_k = \mu_k N$$

are the same for two cases



μ_s = static friction coefficient

μ_k = kinetic friction coefficient

N : normal force

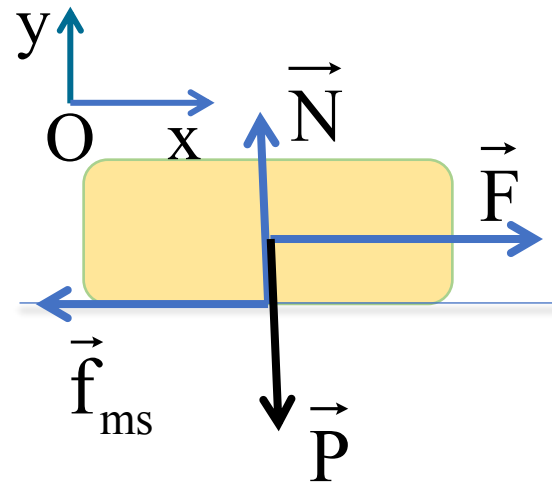
Object at rest: $f_s \leq \mu_s N$

Object moves: $f_k = \mu_k N$

$$\mu_s \geq \mu_k$$

Example 3 (frictional forces)

An object having the weight of 40 N is at rest on a horizontal surface. Pull the object with a horizontal force $\vec{F}$. If the coefficients of static friction and dynamic friction are $\mu_s = 0.5$ and $\mu_k = 0.4$, respectively. What is the magnitude of the frictional force acting on the object? Consider two cases: a) $F = 12$ N. b) $F = 40$ N.



DRAG FORCE

- Drag force appears when an object moves through a fluid such as air or water.
- Drag force or retarding force tends to reduce the speed of the object.
- Drag force depends on the shape of the object, the properties of the fluid, and the speed of the object relative to the fluid.

Unlike ordinary friction, the drag force increases as the speed of the object increases.

At low speeds, the drag force is approximately proportional to the speed of the object

$$F_{\text{drag}} = -kv$$

At higher speeds, the drag force is more nearly proportional to the square of the speed.

$$F_{\text{drag}} = -kv^2$$

k : proportional coefficient

(-) sign means drag force is in the opposite direction of motion

Example

A block rests on an inclined plane surface. The angle of inclination increases until it reaches a critical angle θ_c , after which the block begins to slide. Find the coefficient of static friction μ_s .

$$m\vec{a} = \vec{P} + \vec{N} + \vec{f}_s \quad (1)$$

$$(1) / x : mg \sin \theta - f_s = 0 \quad \Rightarrow f_s = mg \sin \theta$$

$$(1) / y : -mg \cos \theta + N = 0 \quad \Rightarrow N = mg \cos \theta$$

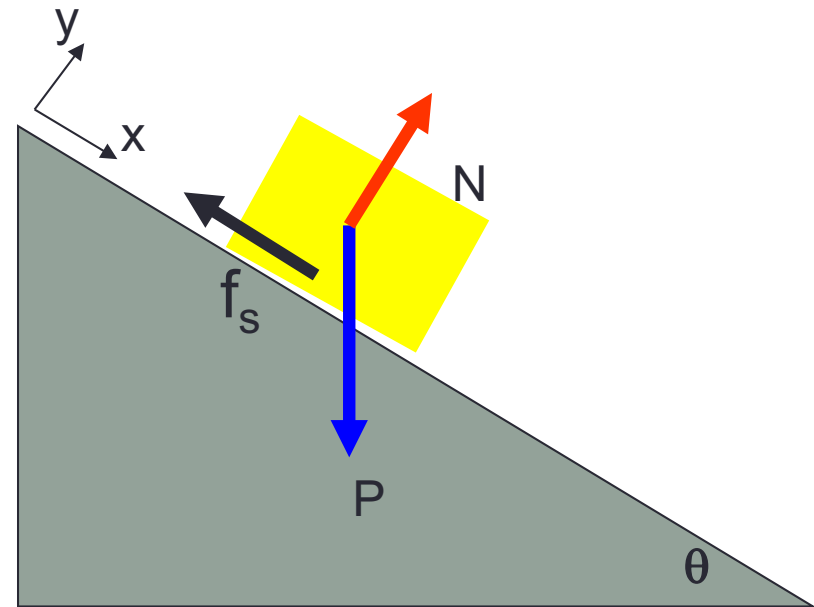
$$f_s \leq (f_{s,\max} = \mu_s N)$$

$$mg \sin \theta \leq \mu_s mg \cos \theta$$

$$\tan \theta \leq \mu_s$$

$$\theta_c = \arctan \mu_s$$

$$\mu_s = \tan \theta_c$$



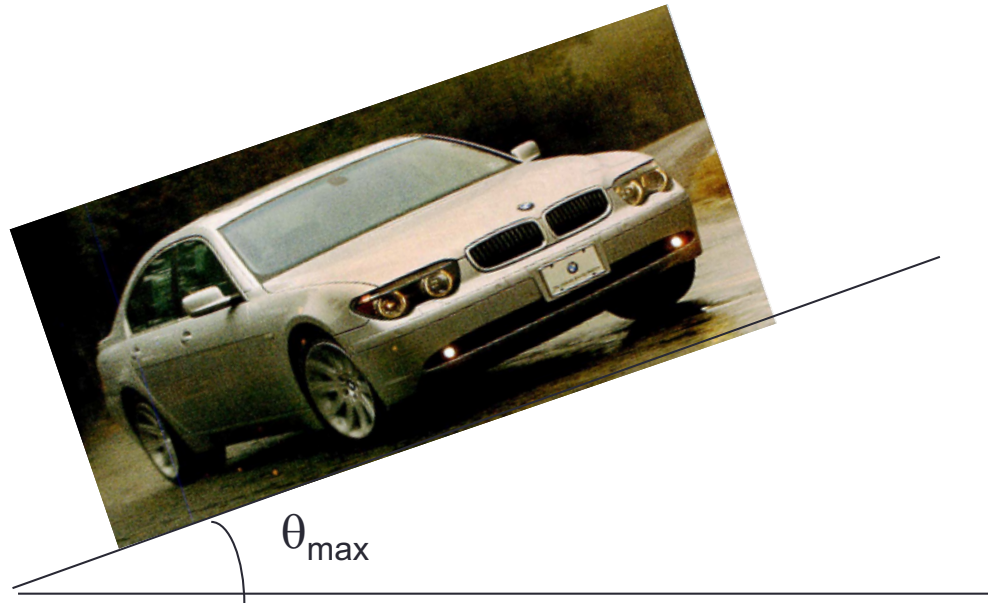
The coefficient of static friction is related to the critical angle θ_c at which an object begins to slip by

$$\mu_s = \tan \theta_c$$

$$\theta_c = \arctan \mu_s$$

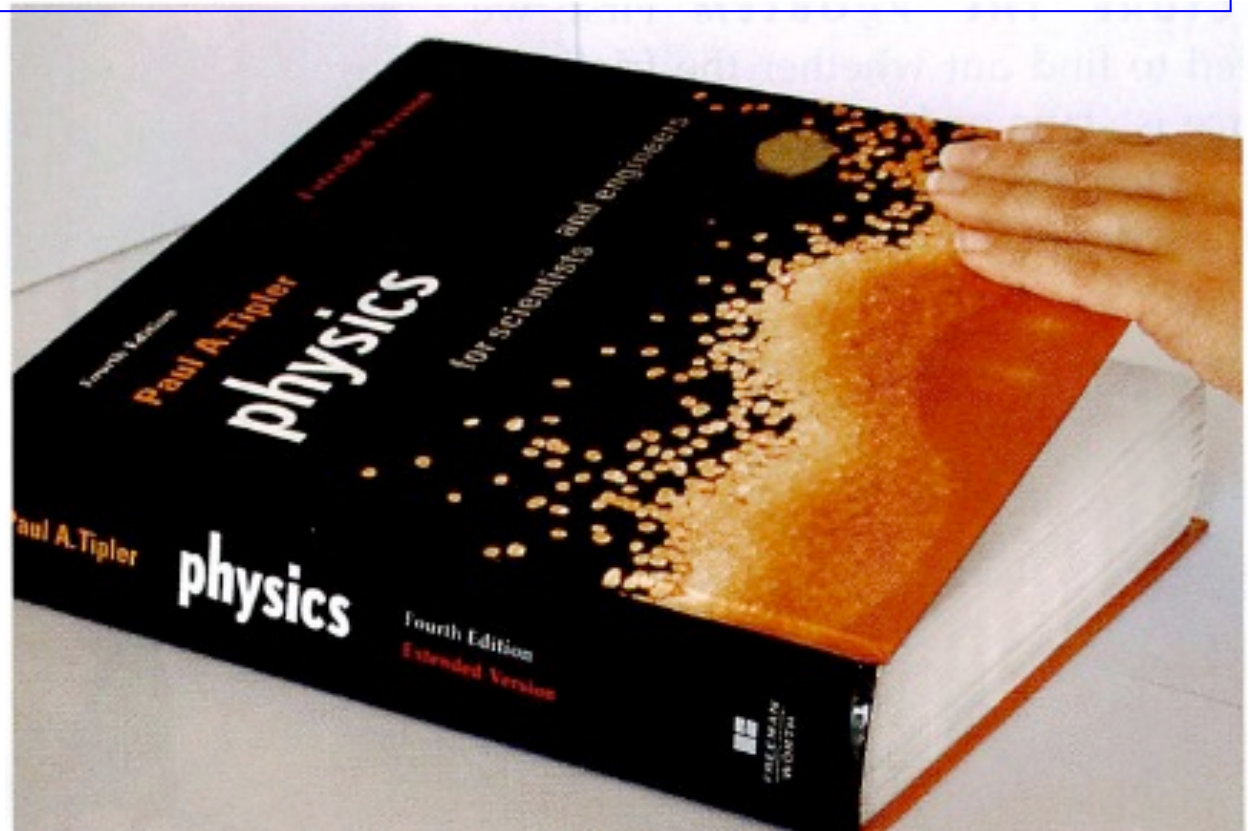
Car on a hill

The coefficient of static friction between a car's tires and the road on a particular day is 0.7. What is the steepest angle of inclination of the road for which the car can be parked with its wheels locked and not slide down the hill? (ANS. 35°)



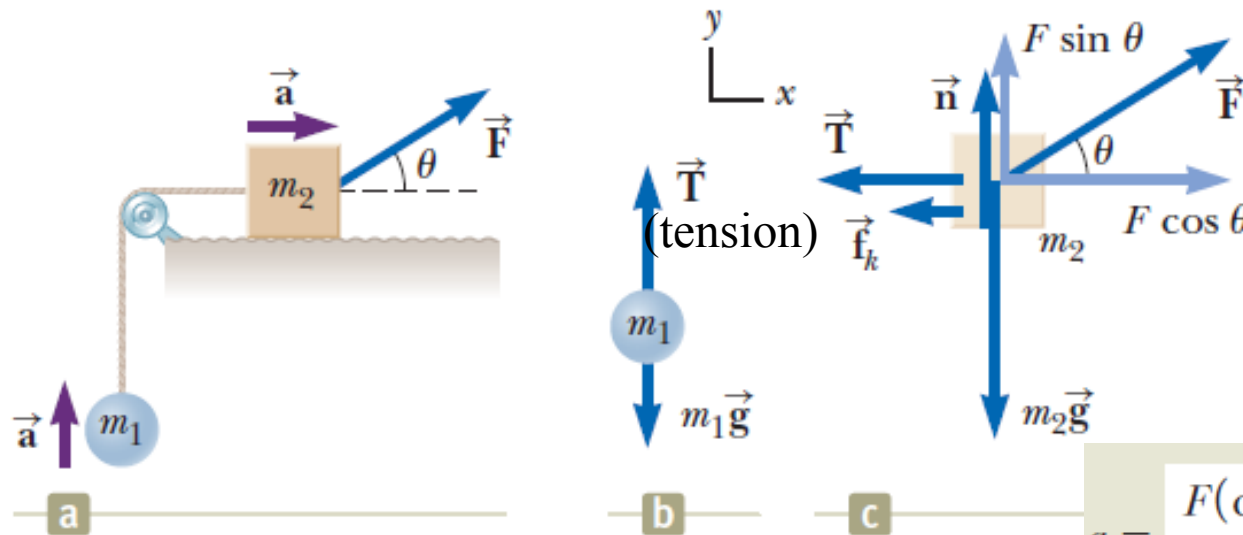
Coin on a book

A hardcover book is resting on a table top with its front cover facing upward. You place a coin on this cover and very slowly open the book until the coin starts to slide. The angle $\theta_{\max}$ is the angle the front cover makes with the horizontal just as the coin starts to slide. Find the coefficient of static friction μ_s between the book cover and the coin in terms of $\theta_{\max}$

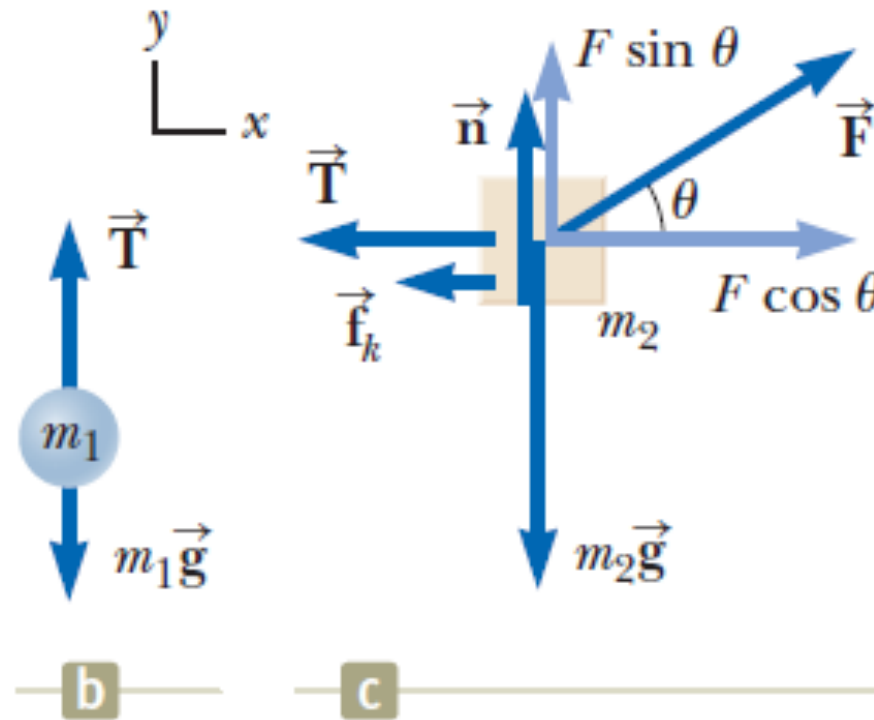
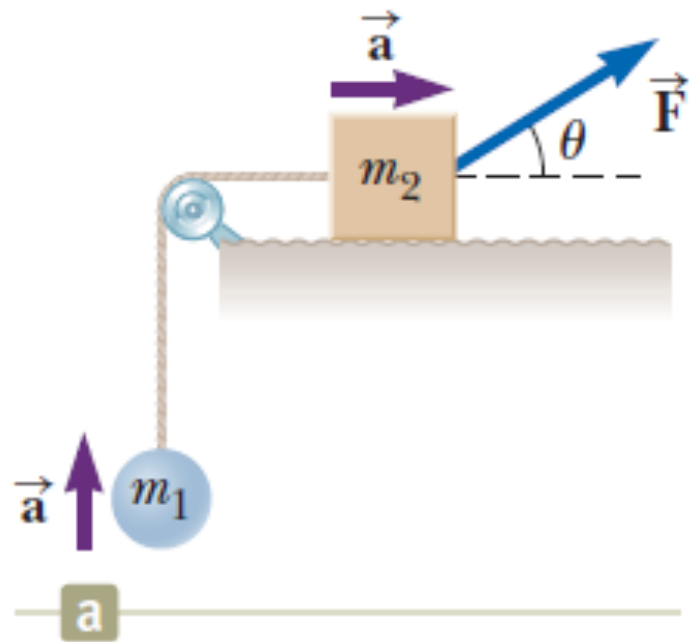


Example 5.13**Acceleration of Two Connected Objects When Friction Is Present**

A block of mass m_2 on a rough, horizontal surface is connected to a ball of mass m_1 by a lightweight cord over a lightweight, frictionless pulley as shown in Figure 5.20a. A force of magnitude F at an angle θ with the horizontal is applied to the block as shown, and the block slides to the right. The coefficient of kinetic friction between the block and surface is μ_k . Determine the magnitude of the acceleration of the two objects.



$$a = \frac{F(\cos \theta + \mu_k \sin \theta) - (m_1 + \mu_k m_2)g}{m_1 + m_2}$$



$$a = \frac{F(\cos \theta + \mu_k \sin \theta) - (m_1 + \mu_k m_2)g}{m_1 + m_2}$$

$$m_1 \vec{a}_1 = \vec{P}_1 + \vec{T} \quad (1)$$

$$m_2 \vec{a}_2 = \vec{P}_2 + \vec{N} + \vec{F} + \vec{f}_k + \vec{T}' \quad (2)$$

$$a_1 = a_2$$

$$T = T'$$

$$(1) / y \quad m_1 a = T - m_1 g$$

$$(2) / x \quad m_2 a = F \cos \theta - f_k - T$$

$$(m_1 + m_2) a = F \cos \theta - m_1 g - f_k$$

$$(3) / y \quad 0 = -m_2 g + N + F \sin \theta$$

$$N = m_2 g - F \sin \theta$$

$$f_k = \mu_k \cdot N = \mu_k m_2 g - \mu_k F \sin \theta$$

$$a = \frac{F \cos \theta - m_1 g - (\mu_k m_2 g - \mu_k F \sin \theta)}{m_1 + m_2}$$

57. An inventive child named Nick wants to reach an apple in a tree without climbing the tree. Sitting in a chair connected to a rope that passes over a frictionless pulley (Fig. P5.57), Nick pulls on the loose end of the rope with such a force that the spring scale reads 250 N. Nick's true weight is 320 N, and the chair weighs 160 N. Nick's feet are not touching the ground. (a) Draw one pair of diagrams showing the forces for Nick and the chair considered as separate systems and another diagram for Nick and the chair considered as one system. (b) Show that the acceleration of the system is *upward* and find its magnitude. (c) Find the force Nick exerts on the chair.



the spring scale reads 250 N.
Nick's true weight is 320N,
the chair weighs 160 N.

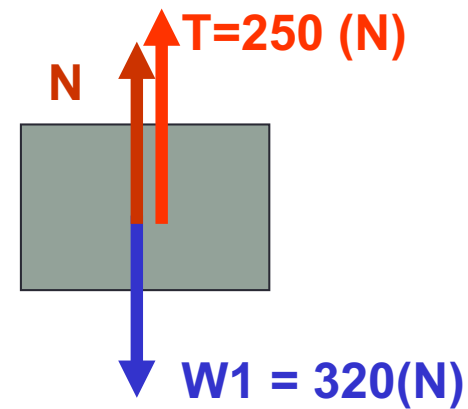
- (a) - Draw free - body diagrams for Nick and the chair considered as separate systems
- diagram for Nick and the chair considered as one system.
- (b) Show that the acceleration of the system is upward and find its magnitude.
- (c) Find the force Nick exerts on the chair.





Nick pulls the cord with a force $\vec{T}'$,
the cord exerts on Nick a force $\vec{T}$.
Forces exerting on the man are :

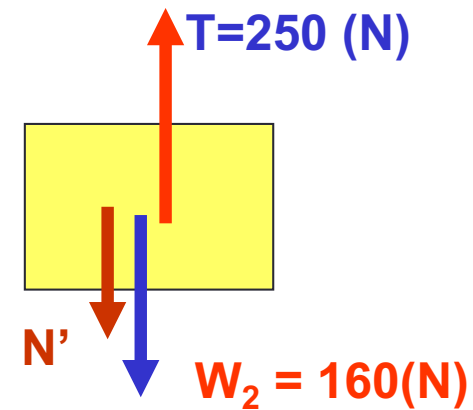
- Weight $\vec{W}_1$ (from the Earth)
- Tension $\vec{T}$ (from the rope)
- Normal force $\vec{N}$ (from the chair)



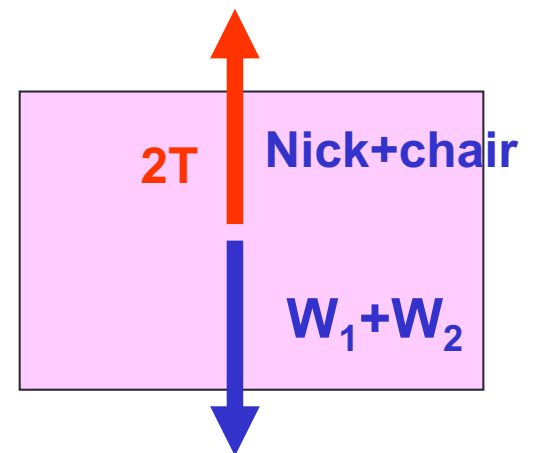
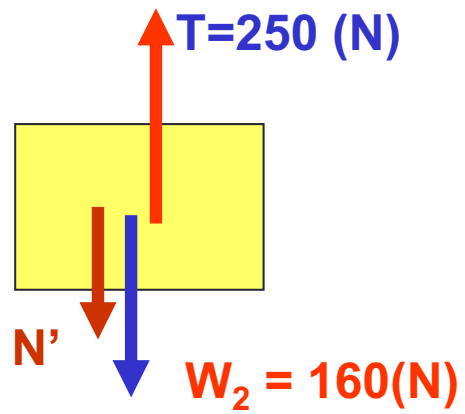
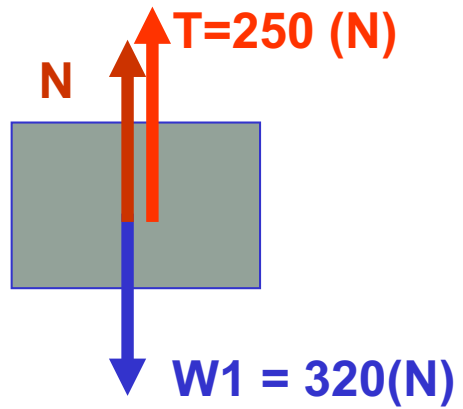
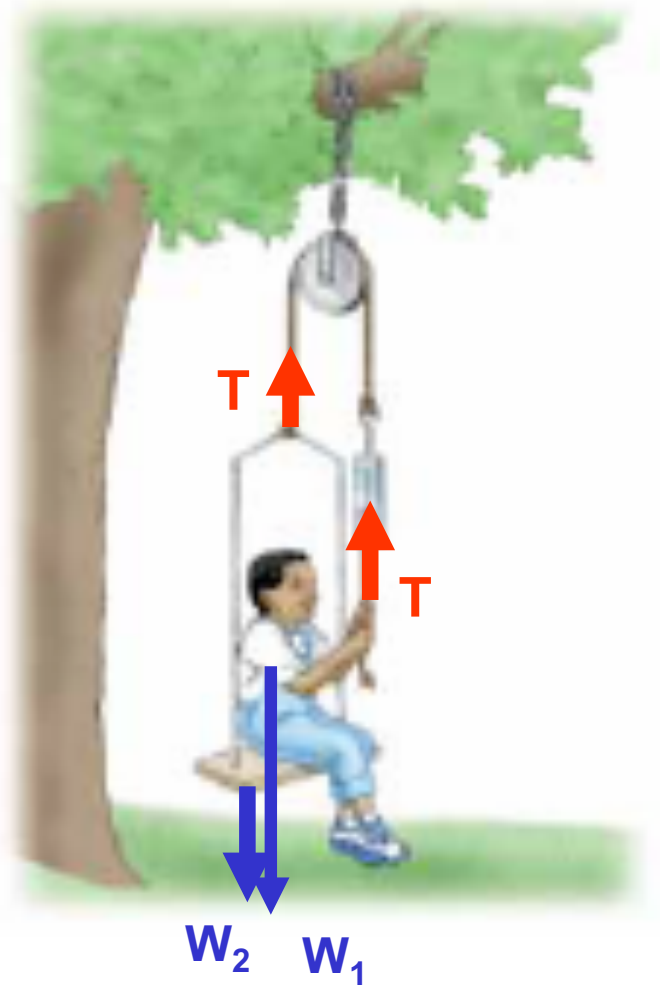
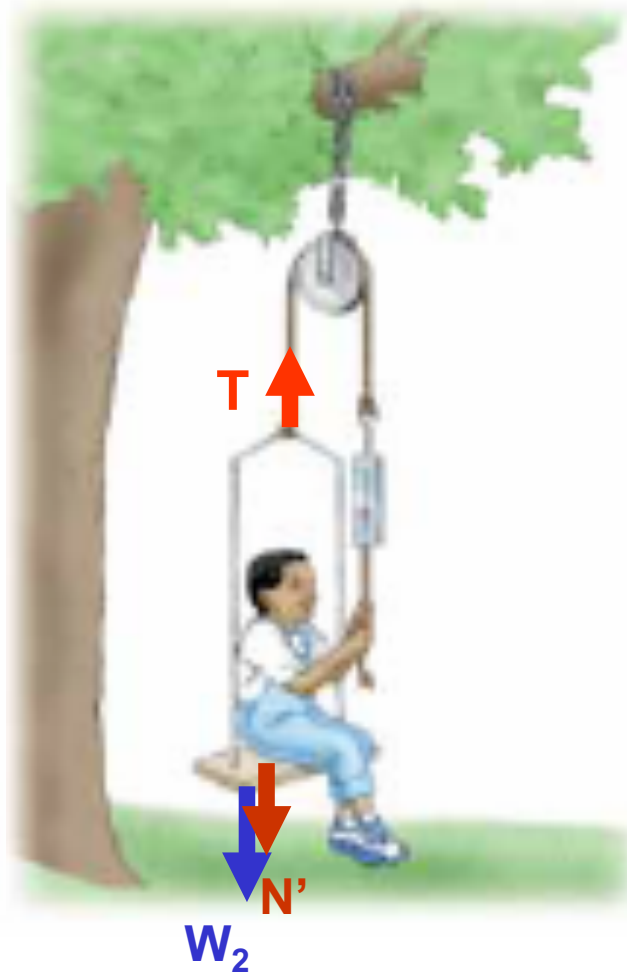
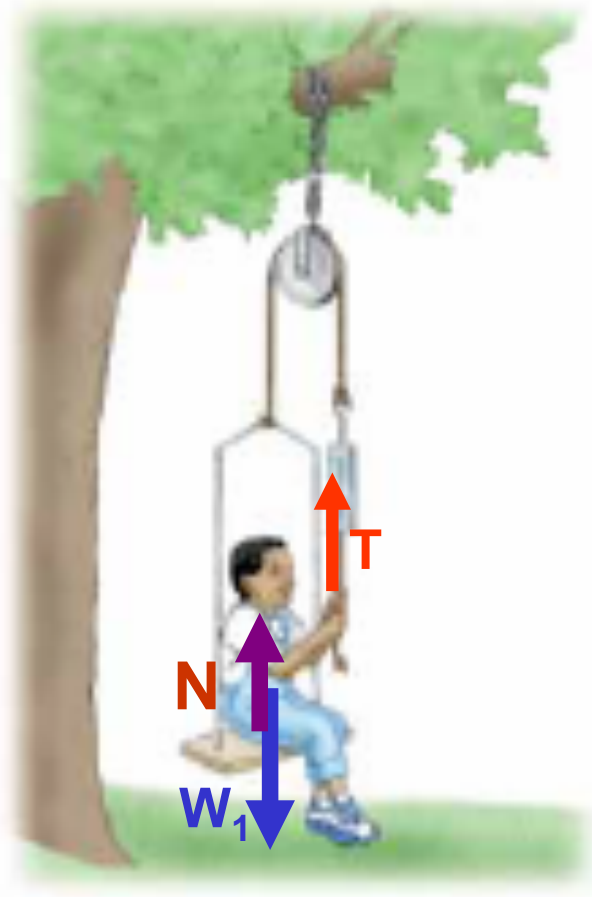
Free-body diagram for Nick

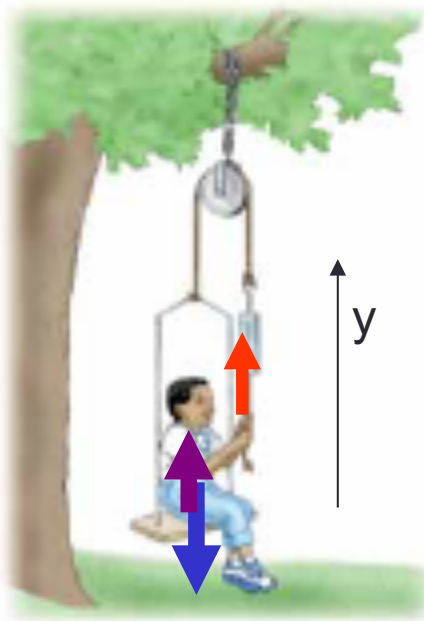


- Forces exerting on the chair
- Weight W_2 (from Earth)
 - Normal force from Nick N'
 - Tension T (from the rope)



Free-body diagram for the chair





Applying the Newton's 2nd law
for Nick and the chair :

$$m_1 \vec{a}_1 = \vec{W}_1 + \vec{N} + \vec{T}(1) \quad (1)/y: m_1 a = -W_1 + N + T(1')$$

$$m_2 \vec{a}_2 = \vec{W}_2 + \vec{N}' + \vec{T}(2)$$

Nick and chair have the same acceleration :

$$\vec{a}_1 = \vec{a}_2 = \vec{a}$$

$$(1) + (2) : (m_1 + m_2) \vec{a} = \vec{W}_1 + \vec{W}_2 + 2\vec{T}(3)$$

Projecting (3) on the upward y axis :

$$(m_1 + m_2) a = -W_1 - W_2 + 2T(3)$$

$$a = \frac{2T - W_1 - W_2}{m_1 + m_2}$$

$$m_1 = W_1/g = 32.7 \text{ kg}, \quad m_2 = W_2/g = 16.3 \text{ kg}$$

$$a = \frac{2 \times 250 - (320 + 160)}{32.7 + 16.3} = 0.408 \text{ m/s}^2 > 0$$

$$(1') m_1 a = -W_1 + N + T$$

$$N = m_1 a + W_1 - T = 32.7 \times 0.408 + 320 - 250 = 83.3 \text{ N}$$

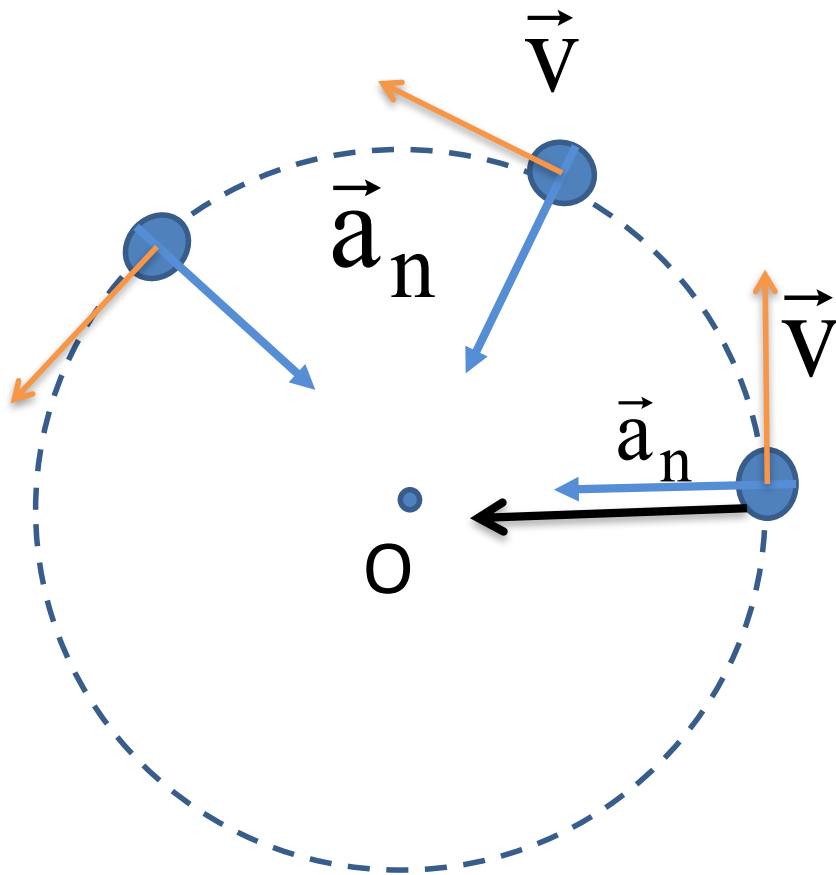


W, N

2.2. CIRCULAR MOTION - APPLICATIONS of NEWTON'S LAWS



6.1 UNIFORM CIRCULAR MOTION



$$\vec{a} = \vec{a}_\tau + \vec{a}_n = \frac{dv}{dt} \vec{\tau} + \frac{v^2}{R} \vec{n}$$

$$\text{uniform} \Rightarrow v = \text{const} \Rightarrow \frac{dv}{dt} = 0 \Rightarrow \vec{a}_\tau = 0$$

$$\vec{a} = \vec{a}_n = \frac{v^2}{R} \vec{n}$$

$$\vec{F} = m\vec{a} = m \frac{v^2}{R} \vec{n}$$

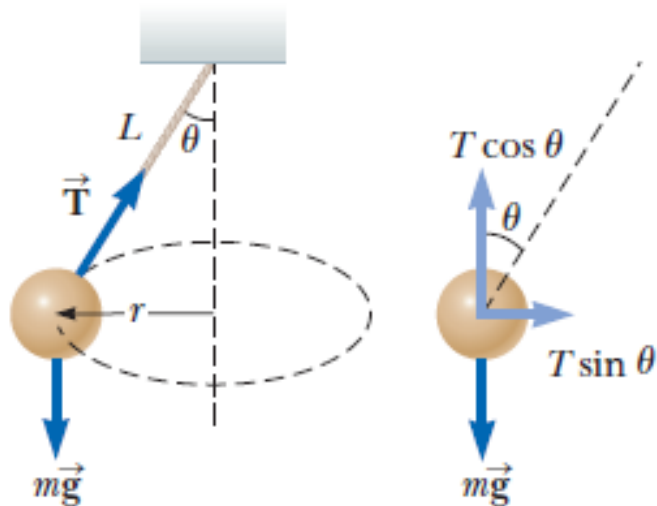
The force causing a centripetal acceleration acts toward the center of the circular path and causes a change in the direction of the velocity vector. (watch video)

<https://www.youtube.com/watch?v=5vtH1uBaoBY>

Example 6.1 The Conical Pendulum

A small ball of mass m is suspended from a string of length L . The ball revolves with constant speed v in a horizontal circle of radius r as shown in Figure 6.3. (Because the string sweeps out the surface of a cone, the system is known as a *conical pendulum*.)

Find an expression for v .



$$m\vec{a} = \vec{P} + \vec{T} \quad (1)$$

$$(1)/_y \quad 0 = -mg + T \cos \theta \Rightarrow T \cos \theta = mg \quad (2)$$

$$(1)/_{\vec{n}} \quad ma_n = T \sin \theta \quad ; \quad a_n = \frac{v^2}{r}$$

$$\left. \begin{array}{l} T \sin \theta = m \frac{v^2}{r} \\ T \cos \theta = mg \end{array} \right\} \Rightarrow \tan \theta = \frac{v^2}{gr}$$

$$\Rightarrow v = \sqrt{gr \tan \theta} \quad ; r = L \sin \theta$$

$$\Rightarrow v = \sqrt{gL \sin \theta \tan \theta}$$

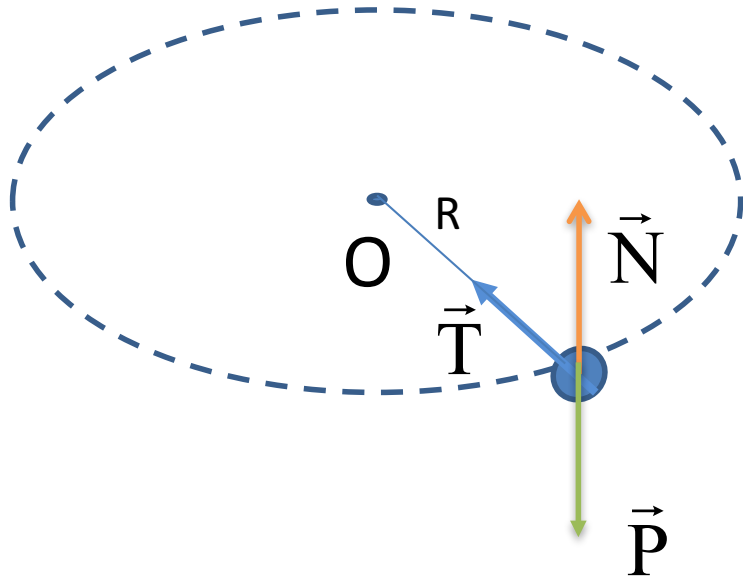
What if $\theta = 90^\circ$?

When $\theta = 90^\circ$ the string is horizontal. the speed v is infinite $\Rightarrow$ the string cannot possibly be horizontal. there would be no vertical component of the force $\vec{T}$ to balance the gravitational force.

Example 6.2 How Fast Can It Spin?

A puck of mass 0.500 kg is attached to the end of a cord 1.50 m long. The puck moves in a horizontal circle as shown in Figure 6.1. If the cord can withstand a maximum tension of 50.0 N, what is the maximum speed at which the puck can move before the cord breaks?

Assume the string remains horizontal during the motion.



$$m\vec{a} = \vec{P} + \vec{N} + \vec{T} \quad (1)$$

$$(1) / \vec{n} \quad ma_n = T \quad (2)$$

$$T = m \frac{v^2}{R} \leq T_{\max}$$

$$v \leq \sqrt{\frac{RT_{\max}}{m}}$$

$$v_{\max} = \sqrt{\frac{RT_{\max}}{m}}$$

$$v_{\max} = \sqrt{\frac{(1.5\text{m}) \times (50\text{N})}{0.5\text{kg}}} = 12.2\text{m/s}$$

Example 6.3 What Is the Maximum Speed of the Car?

A 1500-kg car moving on a flat, horizontal road negotiates a curve as shown in Figure 6.4a. If the radius of the curve is 35.0 m and the coefficient of static friction between the tires and dry pavement is 0.523, find the maximum speed the car can have and still make the turn successfully.



Figure 6.4 (Example 6.3) (a) The force of static friction directed toward the center of the curve keeps the car moving in a circular path. (b) The forces acting on the car.

$$m\vec{a} = \vec{P} + \vec{N} + \vec{f}_s \quad (1)$$

$$(1) / \vec{n} \quad ma_n = f_s \quad (2)$$

$$f_s \leq f_{s.\max}$$

$$f_{s.\max} = \mu_s N = \mu_s mg$$

$$m \frac{v^2}{R} \leq \mu_s mg$$

$$v \leq \sqrt{\mu_s g R}$$

$$v_{\max} = \sqrt{\mu_s g R}$$

$$v_{\max} = \sqrt{0.523 \times (9.8 \text{ m/s}^2) \times (35 \text{ m})}$$
$$= 13.4 \text{ m/s}$$

Example 6.4 The Banked Roadway

A civil engineer wishes to redesign the curved roadway in Example 6.3 in such a way that **a car will not have to rely on friction to round the curve** without skidding. In other words, **a car moving at the designated speed can negotiate the curve even when the road is covered with ice**. Such a ramp is usually *banked*, which means that the roadway is tilted toward the inside of the curve. Suppose the designated speed for the ramp is to be 13.4 m/s (30.0 mi/h) and the radius of the curve is 35.0 m. At what angle should the curve be banked?



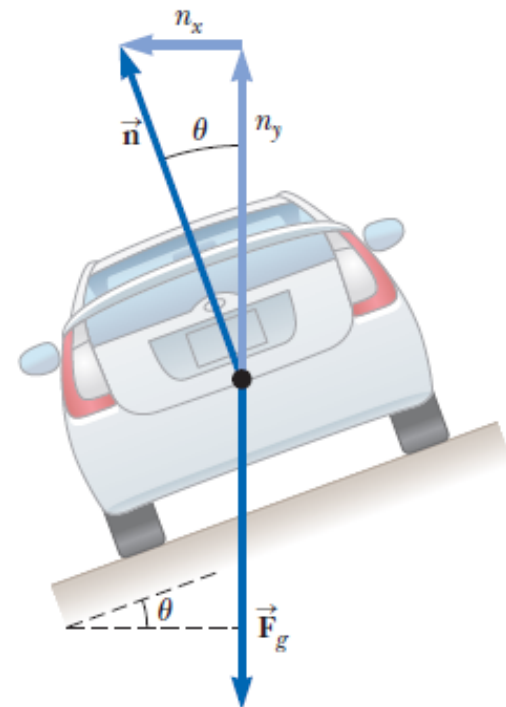
$$\begin{aligned} m\vec{a} &= \vec{P} + \vec{N} & (1) \\ (1) /_{\text{vert.dir.}} & 0 = -mg + N \cos \theta \\ (1) /_{\vec{n}} & ma_n = N \sin \theta & (2) \end{aligned}$$

$$N \sin \theta = m \frac{v^2}{R}$$

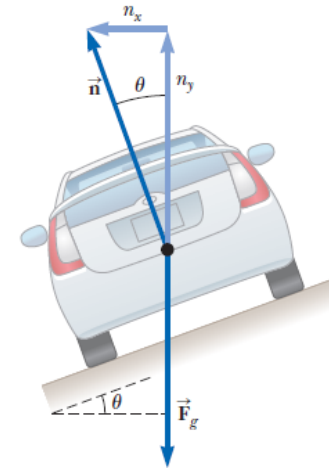
$$N \cos \theta = mg$$

$$\tan \theta = \frac{v^2}{Rg}$$

$$\theta = \arctan \left(\frac{v^2}{Rg} \right)$$



$$\theta = \arctan\left(\frac{v^2}{Rg}\right)$$



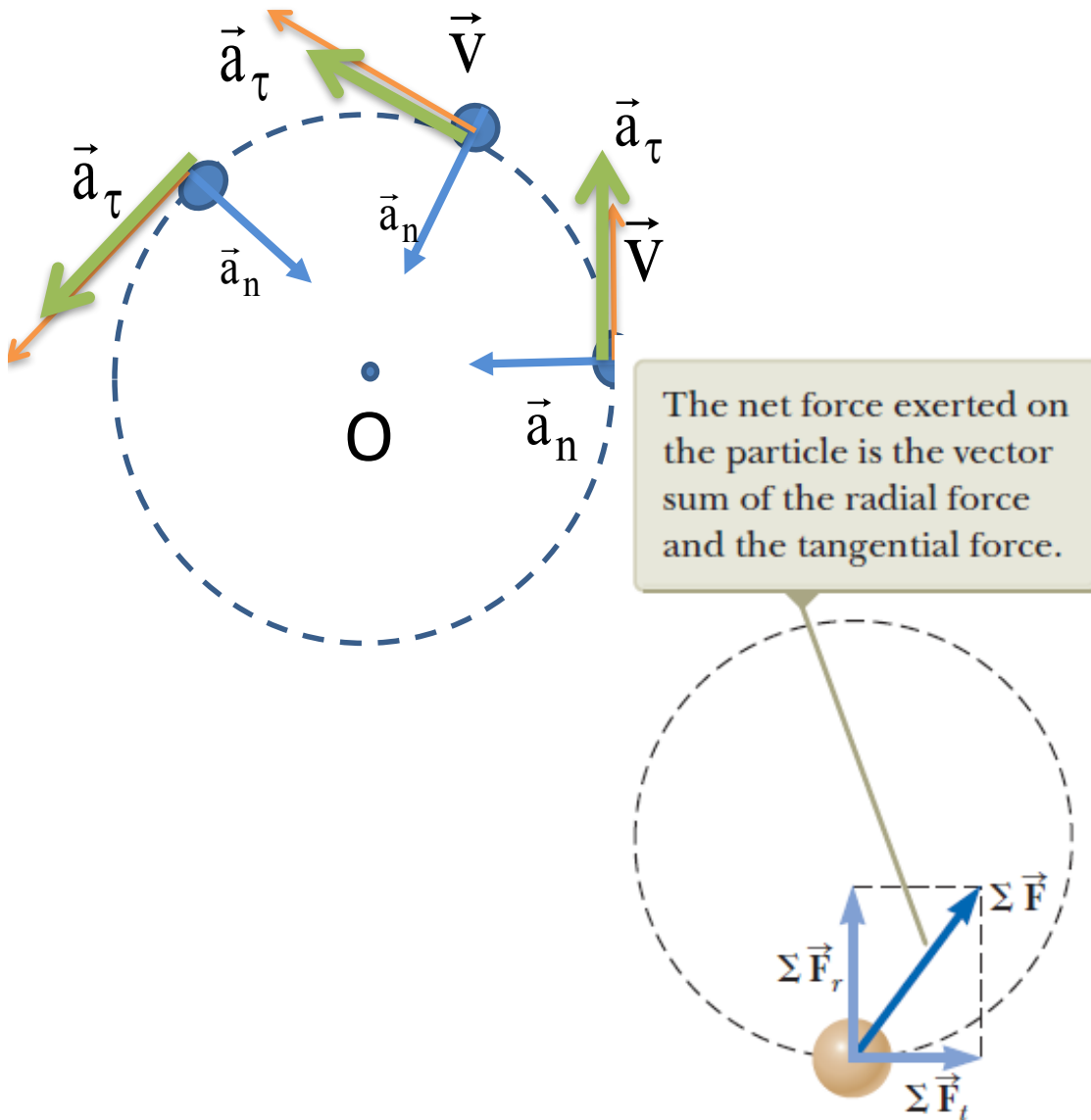
Remark The banking angle θ depends on v and R , but not the mass m ; θ increases with increasing v , and decreases with increasing R .

* When the banking angle, speed, and radius satisfy $\tan \theta = v^2/Rg$, the car rounds the curve smoothly, with no tendency to slide either inward or outward.

* If the car speed is greater than this amount, the road will exert a frictional force down the incline. This force has an inward horizontal component, which provides the additional centripetal force needed to keep the car from moving outward (sliding up the incline).

* If the car speed is less than this amount, the road must exert a frictional force up the incline to keep the car from moving inward (sliding down the incline).

6.2 Non-uniform Circular Motion



$$\vec{a} = \vec{a}_\tau + \vec{a}_n = \frac{dv}{dt} \vec{\tau} + \frac{v^2}{R} \vec{n}$$

$$m\vec{a} = m\vec{a}_\tau + m\vec{a}_n$$

$$\vec{F} = \vec{F}_\tau + \vec{F}_n$$

Example 6.6 Keep Your Eye on the Ball

A small sphere of mass m is attached to the end of a cord of length R and set into motion in a *vertical* circle about a fixed point O as illustrated in Figure. Determine the tangential acceleration of the sphere and the tension in the cord at any instant when the speed of the sphere is v and the cord makes an angle θ with the vertical.

$$m\vec{a} = \vec{P} + \vec{T} \quad (1)$$

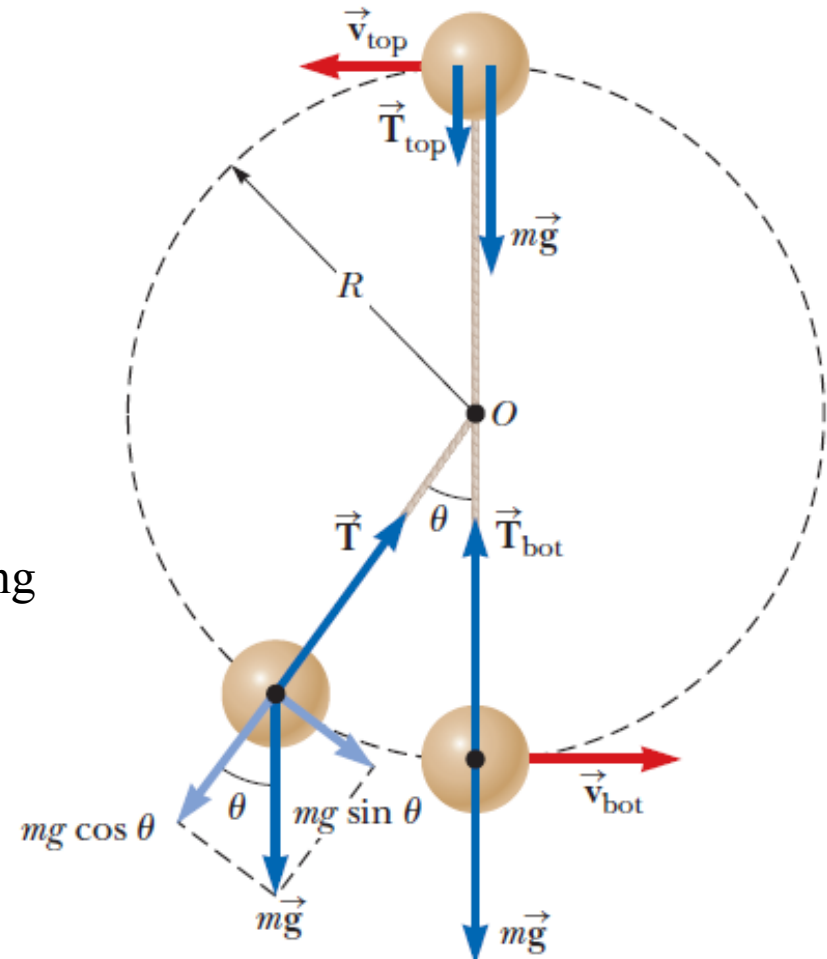
$$ma_n = m \frac{v^2}{R} = -mg \cos \theta + T \quad (2)$$

$$ma_\tau = mg \sin \theta \quad (3)$$

$$a_n = \frac{v^2}{R}; a_\tau = g \sin \theta; T = m \frac{v^2}{R} + mg \cos \theta$$

$$\theta = 0^\circ \quad a_n = \frac{v_{\text{bottom}}^2}{R}; a_\tau = 0; T = m \frac{v_{\text{bottom}}^2}{R} + mg$$

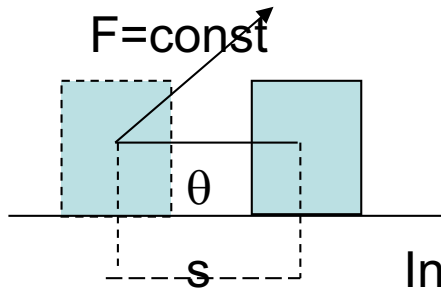
$$\theta = 180^\circ \quad a_n = \frac{v_{\text{top}}^2}{R}; a_\tau = 0; T = m \frac{v_{\text{top}}^2}{R} - mg$$



4.1 Work and Kinetic Energy

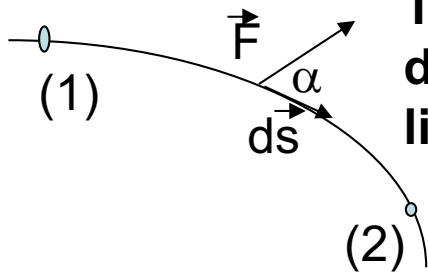
<https://www.youtube.com/watch?v=5-2NLHB4Gxg>

The work done by a **constant force F** on the object when it moves a **straight distance s** is:



$$W = Fs \cos \theta = \vec{F} \cdot \vec{s} \quad (\text{Joule} = \text{Nm} = \text{kg m}^2/\text{s}^2)$$

In general case, the work is not constant, the path is a curve.



The work done by force **F** when the object moves a very small displacement **ds** (we can consider **F** constant and **ds** a straight line):

$$dW = \vec{F} \cdot \vec{ds} = m \frac{d\vec{v}}{dt} \cdot \vec{ds} = m\vec{v} d\vec{v}$$

The work done by force **F** when the object moves from position (1) to (2) is:

$$W = \int_1^2 \vec{F} \cdot \vec{ds} = \int_{\vec{v}_1}^{\vec{v}_2} m\vec{v} d\vec{v} = \frac{1}{2} m\vec{v}_2^2 - \frac{1}{2} m\vec{v}_1^2 \quad \text{We define: Kinetic Energy: } K = \frac{1}{2} mv^2$$

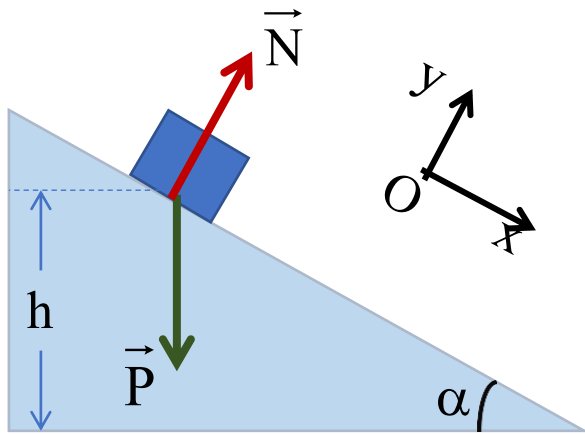
The total work done on a particle is equal to the change in its kinetic energy

$$W = \Delta K = \frac{1}{2} mv_2^2 - \frac{1}{2} mv_1^2 = K_2 - K_1$$

Work-Kinetic Energy Theorem

Example 7

An object of mass m slides without friction from a height h to the bottom of a plane inclined at an angle α with respect to the horizontal direction. Find the expression for the work performed by the gravitational force.



Example 8

A particle moves from position $M (2\text{m}, 0\text{m})$ to position $N (1\text{m}, 2\text{m})$ under the interaction of force $\vec{F} = (3\vec{i} + 4\vec{j})\text{N}$. Calculate the work done by the force.

4.2 Power

$$\text{Power} = \frac{\text{work_done}}{\text{per_unit_of_time}}$$

$$P = \frac{dW}{dt} = \frac{\vec{F} \cdot d\vec{s}}{dt} = \vec{F} \cdot \vec{v} \quad (\text{J/s})$$

$$W = \int dW = \int_{t_1}^{t_2} P dt$$

if $P = \text{const} \Rightarrow W = P\Delta t$ (**P is an average power, denoted by $\langle P \rangle$**)

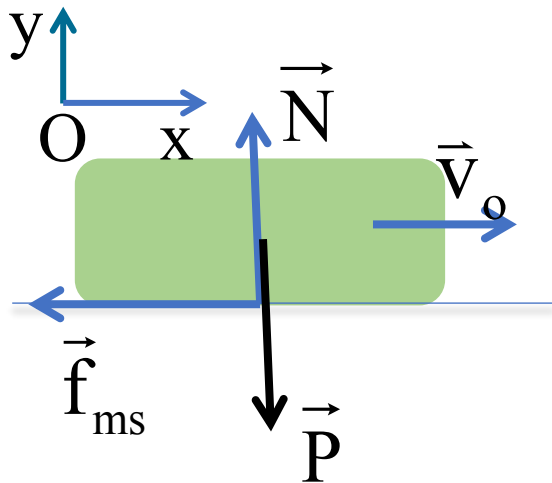
$$W = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2$$

$$\Delta t = \frac{W}{P} = \frac{\frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2}{P}$$

Example 9

A small object with mass of 1kg moves on a horizontal plane with speed $v_0 = 3\text{m/s}$. Knowing that the coefficient of friction between the object and the plane is $k=0.2$, given $g= 10 \text{ m/s}^2$. Find:

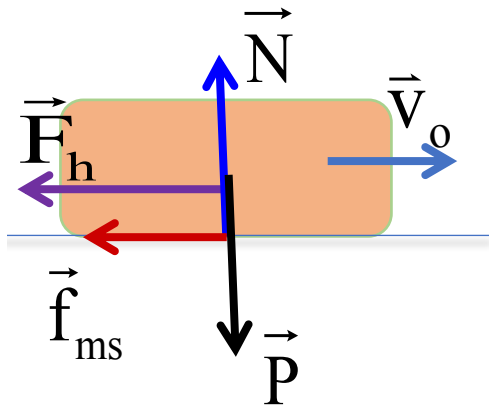
- The power of frictional force at time $t = 1\text{s}$.
- The average power of friction during movement of the object.



Example 10

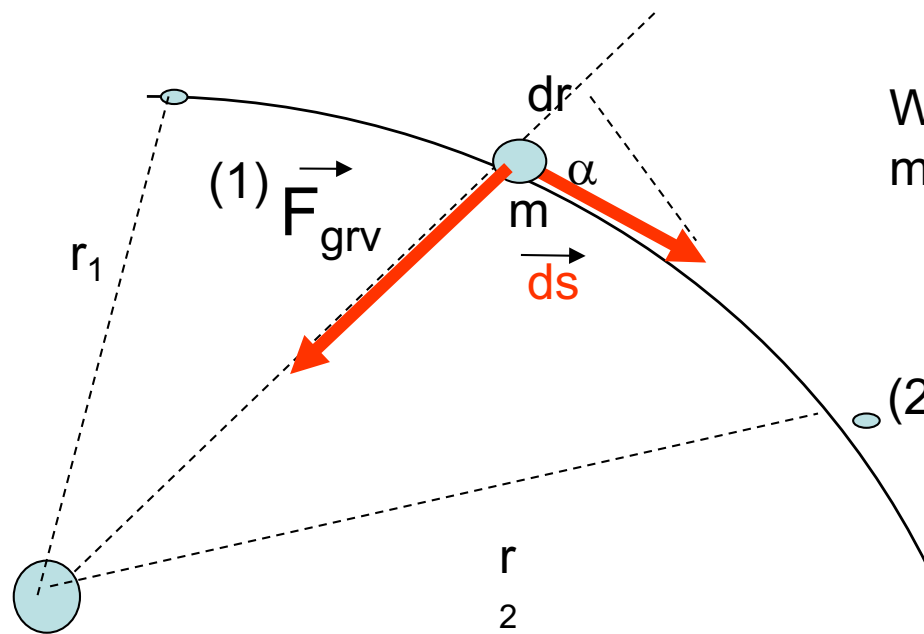
A car with mass $m = 20000$ kg is running at a constant speed on a flat horizontal road and is forced to brake suddenly. Know that the car stops after sliding $s = 45$ m. The braking force of the car is equal to $F_h = 10800$ N. The friction coefficient between the wheel and the road surface is $k = 0.2$. Take the gravitational acceleration $g = 9.8$ m/s². Determine:

- The work of resistant forces acting on the car.
- The speed v_0 of the car before braking.



Conservative & Nonconservative Force

1. **Definition:** A force is Conservative if the work done by the force is independent of the path, it is dependent only on the initial and final positions.
 - Gravity and spring force are conservative forces, while kinetic friction is not.
2. **Work done by Gravitation Force:**



Work done by the grav. force F on object m when it moves a displacement $\underline{ds}$:

$$dW = \vec{F} \cdot \vec{dr} = -F \cdot \underbrace{ds \cdot \cos \alpha}_{dr} = -G \frac{Mm}{r^2} dr$$

(2) Work done by the grav. force F on object m when it moves from (1) to (2)

$$W = \int_{r_1}^{r_2} -G \frac{Mm}{r^2} dr = G \frac{Mm}{r_2} - G \frac{Mm}{r_1}$$

$$G = 6.67 \times 10^{-11} \text{ N.m}^2 / \text{kg}^2$$

Universal Gravitational Constant

+ Work done is independent of the path, but of the initial and final position

⇒ **gravitation force is conservative.**

+ we define a scalar quantity called gravitational potential energy of two object separated by a distance r :

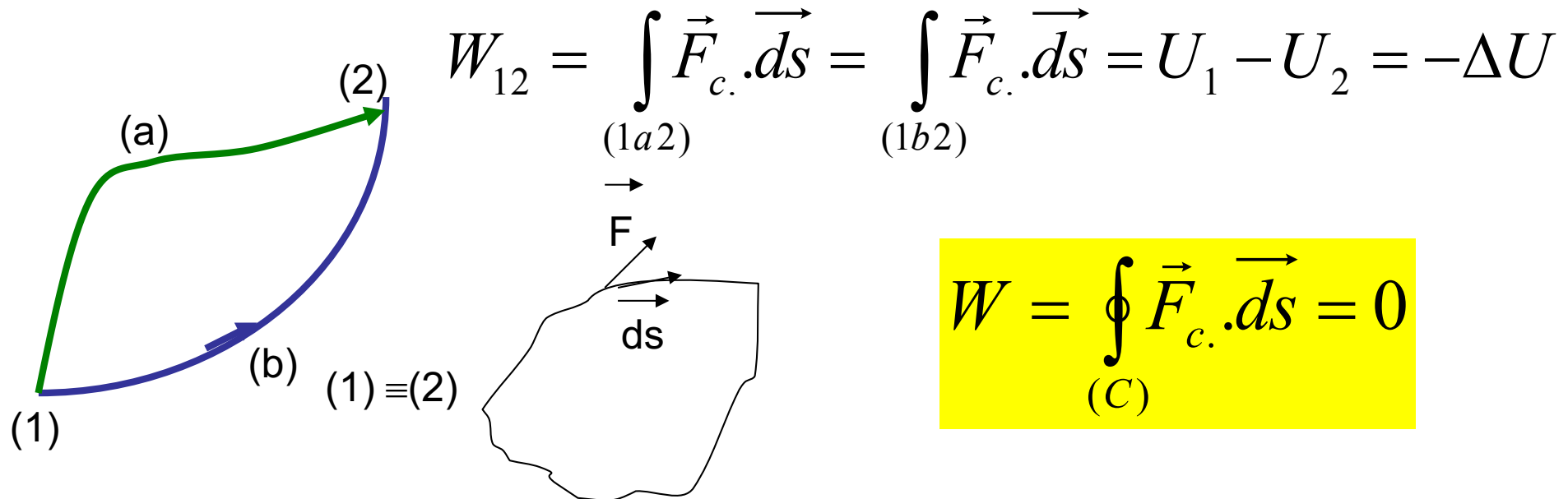
$$U(r) = -G \frac{Mm}{r} + C$$

If we choose $U = 0$ when $r = \infty$, we have $C = 0$, If we choose $U = 0$ on the surface of Earth: $C = GMm/R$

We can write: $W = U_1 - U_2 = -\Delta U$

Conservative Forces

1. The work done by a conservative force on a particle moving between any two points is independent of the path taken by the particle.
2. The work done by a conservative force on a particle moving through any closed path is zero. (A closed path is one in which the beginning and end points are identical.)
3. the work W_c done by a conservative force on an object as the object moves from one position to another is equal to the initial value of the potential energy minus the final value.


$$W_{12} = \int_{(1) \rightarrow (2)} \vec{F}_c \cdot d\vec{s} = \int_{(1b2)} \vec{F}_c \cdot d\vec{s} = U_1 - U_2 = -\Delta U$$
$$W = \oint_C \vec{F}_c \cdot d\vec{s} = 0$$

Mechanical Energy

- If an object is exerted by Conservative Force F_c and Nonconservative Force F_{nc} ,

$$m\vec{a} = \vec{F}_c + \vec{F}_{nc}$$

- from the Work-Kinetic Energy Theorem :
- F_c is conservative:

$$\Delta K = K_2 - K_1 = W_{F_c} + W_{F_{nc}} \quad (1)$$

$$W_{F_c} = -\Delta U = U_1 - U_2 \quad (2)$$

- From (1) and (2):

$$K_2 - K_1 = U_1 - U_2 + W_{F_{nc}}$$

$$(K_2 + U_2) - (K_1 + U_1) = W_{F_{nc}}$$

- Mechanical Energy:
E=K+U

$$\Delta E = E_2 - E_1 = W_{F_{nc}}$$

Conservation of Mechanical Energy

The change in mechanical energy of an object is equal to the work done by nonconservative force on the object as it takes a path from position (1) to (2)

$$\Delta E = E_2 - E_1 = W_{F_{nc}}$$

If $F_{nc} = 0$ or $W_{F_{nc}} = 0 \Rightarrow \Delta E = 0$: $E = \text{const}$
Conservation of Mechanical Energy

Conservative Force & Potential Energy

$$\vec{F} = F_x \vec{i} + F_y \vec{j} + F_z \vec{k} \quad \vec{ds} = dx \vec{i} + dy \vec{j} + dz \vec{k}$$

$$dW = \vec{F}_c \cdot \vec{ds} = F_x dx + F_y dy + F_z dz \quad (1)$$

F is conservative

$$dW = -dU = -\left(\frac{\partial U}{\partial x} dx + \frac{\partial U}{\partial y} dy + \frac{\partial U}{\partial z} dz \right) (2)$$

$$F_x = -\frac{\partial U}{\partial x}; F_y = -\frac{\partial U}{\partial y}; F_z = -\frac{\partial U}{\partial z}$$

$$\vec{F} = -\overrightarrow{\text{grad}} U$$

Gradient Operator

$$\overrightarrow{\text{grad}} = \frac{\partial}{\partial x} \vec{i} + \frac{\partial}{\partial y} \vec{j} + \frac{\partial}{\partial z} \vec{k}$$

Potential Energy and Equilibrium in One Dimension

For a general conservative force **in one dimension**,

$$\vec{F} = F_x \vec{i} \qquad F_x = -\frac{dU}{dx}$$

A particle is in equilibrium if the net force acting on it is zero.
 $dU/dx=0 \Rightarrow U=\text{min or Max}$

Stable Equilibrium: U_{min}

Unstable Equilibrium: U_{max}

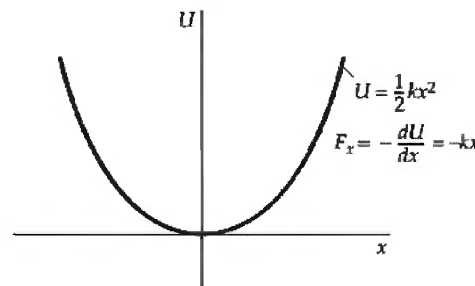


Figure 6-26 Plot of the potential-energy function U versus x for an object on a spring. A minimum in a potential energy curve is a point of stable equilibrium. Displacement in either direction results in a force directed toward the equilibrium position.

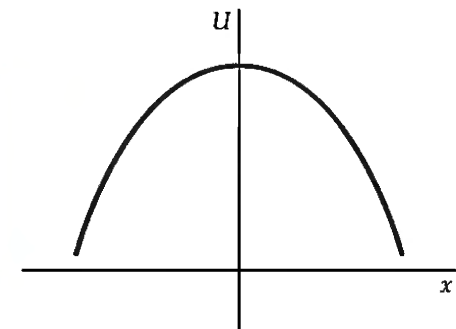
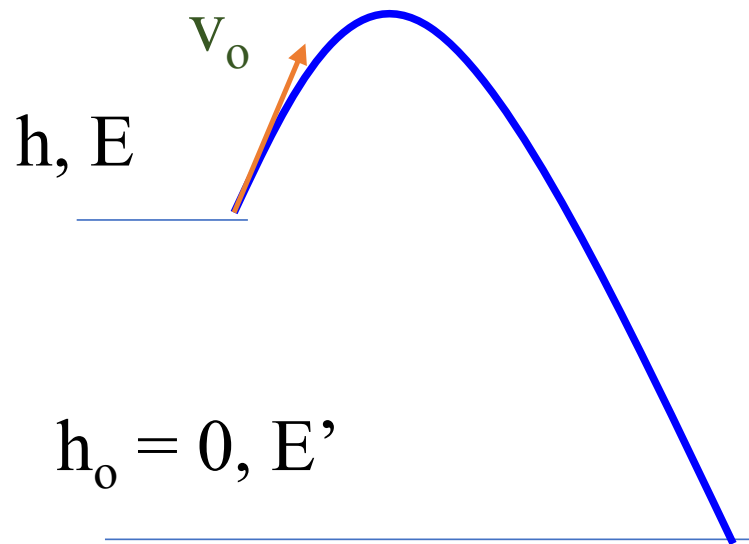


Figure 6-27 A particle at $x = 0$ on this potential-energy curve will be in unstable equilibrium because a displacement in either direction results in a force directed away from the equilibrium position.

Example 11

From a height $h = 20$ m, a stone of mass $m = 200$ g is thrown with an initial speed of $v_0 = 18$ m/s in a direction inclined to the horizontal. When it hits the ground, the stone has a speed of $v' = 24$ m/s. Take the gravitational acceleration $g = 9.8$ m/s². Calculate the work done by the air resistance acting on the stone?



Example 12

An object with $m = 2\text{kg}$ moving horizontally without friction is subjected to a force $\vec{F} = (2 - 6x)\vec{i}$ (N). Choose the origin of the object's potential energy at $x = 0$ to be equal to 0.

- a) Determine the expression for the object's potential energy.
- b) If the particle has a speed $v = 5 \text{ m/s}$ at position 1.0 m , what is the speed v' of the object at position 0.0 m ?

Example 13

An object of mass m slides from a height h down an inclined plane at an angle α with respect to the horizontal. The coefficient of friction between the object and the surface is k . Calculate:

- a. Speed v of the object at the base of the inclined plane.
- b. Frictional force is applied throughout the process of moving.

Work of the fictional force?

Problem 1

1. A particle moves along the x axis under the influence of a stationary object.

The net force on the particle is given by $F = (8\text{N/m}^3)x^3$.

If the potential energy is taken to be zero for $x = 0$ then the potential energy is given by :

A. $(2\text{ J/m}^4)x^4$ B. $(-2\text{J/m}^4)x^4$

C. (24 J/m^2x^2) D. $(-24\text{ J/m}^2)x^2$ E. $5\text{ J} - (2\text{ J/m}^4)x^4$

$$F = -\frac{dU}{dx}$$

$$\Rightarrow -dU = Fdx$$

$$-\int_{U(x=0)=0}^{U(x)} dU = \int_0^x 8x^3 dx$$

$$-U = 2x^4$$

$$1\text{ J} = 1\text{ Nm}$$

ans : B

Problem 2

A 0.20 - kg particle moves along the x axis under the influence of a stationary object.

The potential energy is given by

$$U(x) = (8.0\text{J/m}^2)x^2 + (2.0\text{J/m}^4)x^4,$$

where x is in coordinate of the particle.

If the particle has a speed of 5.0m/s when it is at x = 1.0m,

its speed when it is at the origin is :

- A. 0 B. 2.5m/s C. 5.7m/s
D. 7.9m/s E. 11m/s

ans : E

$$\text{Given } U = 8x^2 + 2x^4$$

$$\text{PE at } x = 1\text{m} : U = 8 \times 1^2 + 2 \times 1^4 = 10\text{J}$$

$$\text{KE at } x = 1\text{m} : T = \frac{1}{2}mv^2 = \frac{1}{2} \times 0.2 \times 5^2 = 2.5\text{J}$$

$$\text{Mechanical Energy at } x = 1\text{m} : E = U + T = 12.5\text{J}$$

The force exerting on the object is conservative force,

so there is **conservation** of Mech. Energy.

$$E|_{x=0} = E|_{x=1\text{m}} = E$$

$$E|_{x=0} = U|_{x=0} + T|_{x=0} = 0 + \frac{1}{2}mv_0^2$$

$$v_0 = \sqrt{\frac{2E}{m}} = \sqrt{\frac{2 \times 12.5}{0.2}} = 11.2\text{m/s}$$

Problem 3

The potential energy of a particle moving along the x axis is given by

$$U(x) = (8.0\text{J/m}^2)x^2 + (2.0\text{J/m}^4)x^4.$$

If the total mechanical energy is 9.0 J, the limits of motion are :

- A. -0.96 m; +0.96m
- B. -2.2m; +2.2m
- C. -1.6m; +1.6m
- D. -0.96 m; +2.2m
- E. -0.96 m; +1.6m

Force is conservative because there is P.E.

Under the action of the conservative force,
there is Conservation of Mech. Energy :

$$E = U + T = \text{const}$$

The limits of motion are the positions where the velocities are zero.

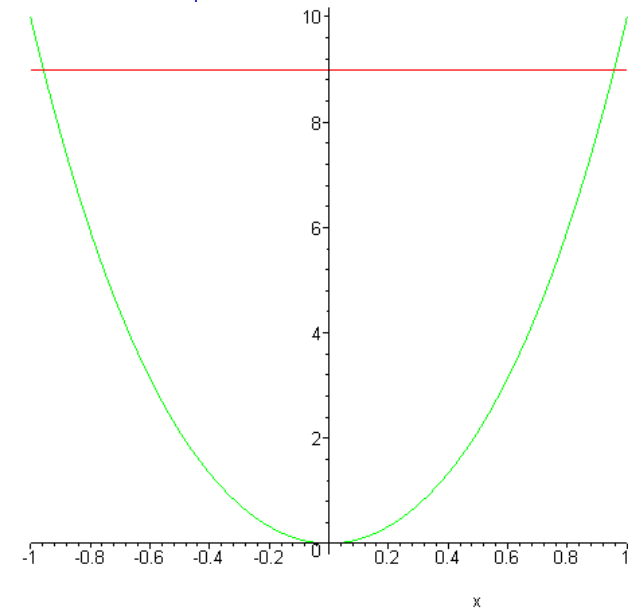
=> kinetic energy is zero.

$$8x_0^2 + 2x_0^4 + 0 = E$$

$$E = 9(\text{J})$$

$$2x_0^4 + 8x_0^2 - 9 = 0$$

$$x_0 = \pm 0.96\text{m}$$



Plot U(x)

ans : A

Problem 4

The potential energy of a particle moving along the x axis is given by

$$U(x) = (8.0\text{J/m}^2)x^2 + (2.0\text{J/m}^4)x^4.$$

Find a) Force b) Equilibrium position c) Is the equilibrium stable?

$$a) F_x = -\frac{dU}{dx} = -16x - 8x^3$$

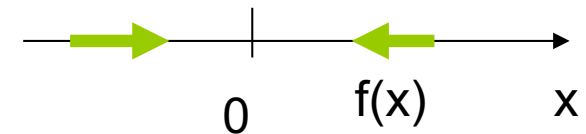
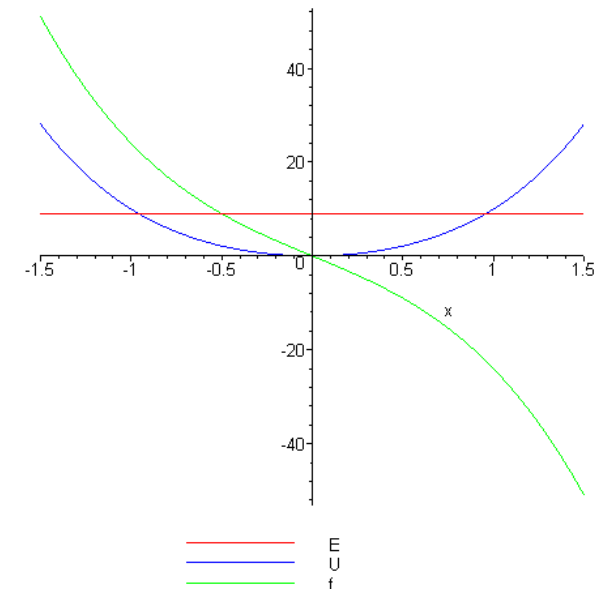
$$b) \frac{dU}{dx} = 8x^3 + 16x = 0 \Rightarrow x = 0$$

$$c) \frac{d^2U}{dx^2} = 24x^2 + 16$$

$$\text{at } x = 0, \frac{d^2U}{dx^2} = 16 > 0$$

Conclusion : $x = 0$ is the equilibrium position.

This equilibrium is stable



9 • The 3 - kg object in Figure 7 -18 is released from rest at a height of 5 m on a curved frictionless ramp. At the foot of the ramp is a spring of force constant $k = 400 \text{ N/m}$. The object slides down the ramp and into the spring, compressing it a distance x before coming momentarily to rest. (a) Find x . (b) What happens to the object after it comes to rest?

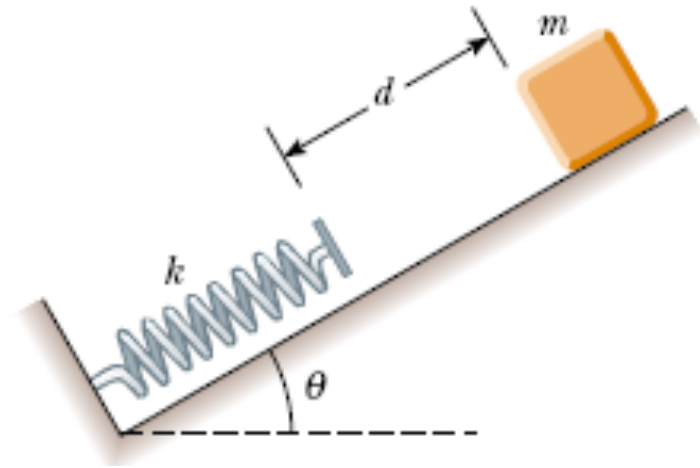


9. (a) 0.858 m; (b) The spring will accelerate the mass and it will then retrace the path, rising to a height of 5 m.

$$mgh = \frac{1}{2} kx^2$$

$$x = \sqrt{\frac{2mgh}{k}} = 0.587 \text{ m}$$

10. An object of mass m starts from rest and slides a distance d down a frictionless incline of angle θ . While sliding, it contacts an unstressed spring of negligible mass as shown in Figure P8.10. The object slides an additional distance x as it is brought momentarily to rest by compression of the spring (of force constant k). Find the initial separation d between object and spring.



$$E_i = E_f$$

$$mgd \sin \theta = -mgx \sin \theta + \frac{1}{2} kx^2$$

$$d = \frac{-mgx \sin \theta + \frac{1}{2} kx^2}{mg \sin \theta} = \frac{1}{2} \frac{kx^2}{mg \sin \theta} - x$$